

Khulna University of Engineering & Technology
Department of Building Engineering and Construction Management
B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2022
MATH 2123
(Mathematics - III)

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⊗ Time: 3 hrs

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section - A

1. (a) A space curve is given by the position vector $2\cos t\hat{i} + 2\sin t\hat{j} + \hat{k}$. Find the unit tangent vector $\hat{T}$, unit normal vector $\hat{N}$, radius of curvature, and normal plane at $t = \frac{\pi}{2}$. Also, draw the space curve roughly. (28)

- (b) How can you identify that a vector field is both solenoidal and rotational? Explain it and give a figure. (07)

2. (a) Find the value of a, b, c so that the directional derivative of $f = ayz^2 + bxz + cx^2y^3$ at $(2, -1, 1)$ has a maximum of magnitude 64 in the direction parallel to y -axis. Hence find the directional derivative of $f = c$ along x -axis at the point $(1, -1, 2)$. (12)

- (b) Define solenoidal vector and conservative vector field. Show that $\vec{A} = (2x^2 + 8xy^2z)\hat{i} + (3x^3y - 3xy)\hat{j} - (4y^2z^2 + 2x^2z)\hat{k}$ is not solenoidal but $\vec{B} = xyz^2\vec{A}$ is solenoidal. (11)

- (c) Define vector and scalar field with example. Find the work done in moving a particle in a force field given by $\vec{F} = 3xy\hat{i} - 5z\hat{j} + 10x\hat{k}$ along the curve $x = t^2 + 1, y = 2t^2, z = t^3; 1 \leq t \leq 2$. (12)

3. (a) Evaluate $\iint_S \vec{F} \cdot \hat{n} ds$, where S is the surface $2x + 2y + z = 4$ located in the first octant cutting by the planes $x = 0, y = 0, z = 0$. Given $\vec{F} = (3z)\hat{i} - 2xy\hat{j} - 4x\hat{k}$. (12)

- (b) If $\vec{F} = (2x + y)\hat{i} + (3y - x)\hat{j}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is the curve in the xy plane consisting of the straight lines from $(0, 0)$ to $(2, 0)$ then to $(3, 2)$. (13)

- (c) Evaluate $\iiint_V \nabla \cdot \vec{F} dv$, where $\vec{F} = (2x^3 - 3z)\hat{i} - 2xy\hat{j} - 4x\hat{k}$ and V is the closed region bounded by the planes $x = 1, y = 1, z = 1$, & $x = 3, y = 3$, and $z = 3$. (10)

4. (a) Reduce the following matrix to its echelon form, canonical form.
- Also find its rank. Here $A = \begin{bmatrix} 0 & 1 & 0 & -2 \\ -4 & 2 & 1 & 1 \\ 2 & 0 & 1 & 0 \end{bmatrix}$
- (b) What is the value of k for which the following system of linear equations has (i) a unique solution, (ii) many solutions and (iii) no solution? If possible, find the unique solution for that particular value of k .
- $$\begin{bmatrix} 2 & 3 & k \\ 1 & 6 & 2k+3 \\ 2 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Section – B

5. (a) Define Laplace transform. Find the Laplace transform of $t^3 \sin t$ and hence, find the (12)

value of $\int_0^{\infty} t^3 e^{-t} \sin t dt \rightarrow L\{t^3 \sin t\}$ at $s=1$
 $\downarrow$
 $L\{f(t)\} = F(s)$ at $s=1$ value
 $\therefore$ value of $F(s) = \frac{6}{(s^2+1)^2}$ at $s=1$
 $= \frac{6}{(1^2+1)^2} = \frac{6}{4} = \frac{3}{2}$

10. (b) Let $F(t) = \begin{cases} 3t & ; 0 < t < 2 \\ 6 & ; 2 < t < 4 \end{cases}$

Where $F(t)$ has period 4. Find $L\{F(t)\}$.

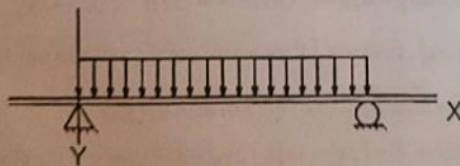
- (c) State convolution theorem. Apply this theorem to find $L^{-1}\left\{\frac{1}{S^2(S+1)^2}\right\}$ $f(s) = \frac{1}{(s+1)^2}$

Find the inverse Laplace transform of $\frac{S+4}{S(S-1)(S^2+4)}$ (12)

Solve the Differential equation $y''(t) + 4y'(t) + 4y(t) = 4e^{-2t}$, $y(0) = -1$, $y'(0) = 4$, (12)
with the help of Laplace transform.

Using Laplace transform solve the integral equation $F(t) = 1 + 2 \int_0^t e^{-x} F(t-x) dx$ (11)

7. (a) A beam is hinged at its ends $x = 0$ and $x = l$, which carries a uniform load w_0 per unit length as shown in the figure below. Find the deflection at any point.



- (b) Define orthogonal matrix. If $(l, m, n), r = 1, 2, 3$ be the direction cosine of three mutually perpendicular lines referred to an orthogonal coordinates system then prove

that $A = \begin{bmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{bmatrix}$ is orthogonal.

- (c) Find the eigen values and the eigen vector corresponding to the largest eigen value of (10)

$$\Lambda, A = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$$

8. (a) Given $2x - y + 3z = 0$, $3x + 2y + z = 0$, $x + 3y - 2z = 0$. If A is the coefficient matrix (20)
and x will be the solution of the above system of linear equations then find the
solution. Again find the solution of the following system for $(BA)X = 0$, where the

transition matrix $B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

Is the solution space changed? Explain.

(15)

- (b) Find A^{-1} where $A = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$. Given $Ax = H$

Where $H = [1 \ 0 \ 1]^T$ then find the value of x .